Tunnels and Underground Structures, Zhao, Shirlaw & Krishnan (eds)
© 2000 Taylor & Francis, ISBN 90 5809 171 6

Tunnelling and underground projects in Singapore

R. Krishnan
Land Transport Authority, Singapore

ABSTRACT: A brief review of the tunnelling and underground works carried out todate in Singapore is presented together with the main soil formation in which they are carried out. A listing of the equipment used for tunnelling works is also provided. Subsurface works now being undertaken including the 20 km North East Line, Changi Airport Line, Deep Tunnel Sewerage System and underground cavern development follows. A brief look at the future works then follows.

1 INTRODUCTION

After several years of study the Government of Singapore in 1982 decided to implement a comprehensive mass rapid transit (MRT) system. This marked the beginning of major tunnelling and underground works in Singapore

Prior to this tunnelling had been carried out to some limited extent. These include a number of small diameter 'mole' machines for the laying of sewers by pipe jacking method. In 1983 an earth pressure balance machine (EPBM) was used to construct a 3.7 m diameter tunnel approximately 3 km long to accommodate a 3.3 m diameter sewage pipeline.

Initially, the tunnelling works for the MRT line was mainly confined to the heavily built up downtown area. However, due to the escalating cost of land and the necessity to conserve surface land due to the tremendous development potential around the MRT stations the trend has been to move MRT lines, even in the outlying areas, below ground. This has opened up tremendous scope for tunnelling and underground works in Singapore.

2 GEOLOGICAL PROFILE

The geological formations of Singapore encountered during tunnelling and excavation works can be broadly covered into 4 main types:

Kallang formation - These are recent deposits found in river valleys, river mouths and coastal environment. Marine clay and peaty soils form the most distinctive soft deposits of this formation. Tunnelling through this material tests the limits of soft soil tunnelling and is often carried out after extensive soil treatment.

Old Alluvium – This material is of relatively young geological age and occupies a large proportion of the surface area, especially in the eastern part of Singapore. It mainly consists of medium dense to very dense clayey coarse sand and fine gravel. Relatively easier to work with TBMS, especially in the unweathered layers and the material is generally abrasive.

Jurong Sedimentary – These are sedimentary series consisting of heavily folded interbedding of sandstone, siltstone and mudstone weathered to varying degrees. Uncertainty of the material at fracture and fault zones with loose friable material and the rapidly changing layers of

materials varying from slightly weathered sandstone to highly weathered/residual clayey silt and sandy clay have posed major challenges in tunnelling through the material.

Bukit Timah granite – Igneous rocks; fresh at depth and completely weathered approaching the ground surface, producing silty clay. The fresh granite rock has a very high strength (average 190MPa). The main challenge has been tunnelling through mixed face of softer material with the hard fresh granite intruding into the invert and lower face of the TBM. This cause high torque and temperatures to build up together with difficulties in steering the machine.

3 COMPLETED MAJOR TUNNELING PROJECTS

3.1 *Tunnels for the transit system*

Tunnels for the MRT system constitutes the majority of tunnels constructed todate in Singapore. Tunnelling first started in 1983 with the commencement of the first two phases of the MRT system. 19 km of the system of those phases were underground with 11 km of it in twin tunnel; 5.2m internal diameter. 15 of the stations are also underground.

The drives carried out in 38 separate tunnel drives with a wide variety of equipment used to cater to the varying ground conditions. These include Greathead type shields with backhoe or milling head for mechanised excavation to full faced TBMs with hydraulic face gates and EPB machines. (Figure 1)

The whole of the 67km of line and 42 stations under first two phases was carried out between 1983 and 1990. Though the system has been extended to 83 km with 48 stations, the extension is all above ground as thus there was no mechanised tunnelling carried out between 1987 and 1998 till the commencement of the North East Line.

3.2 *Cable Tunnel to Pulau Seraya*

The undersea tunnel, to carry electric cables from the power station on Pulau Seraya to the mainland was constructed in 1988. The route was 2.6 km overall with 1.8 km of it below the sea bed. The tunnel is rectangular, 6.5 m wide and 3.7 m high and laid as a submerged tube. Each section of the tube was made up of 28 numbers of 3.5 m long match cast segments and a special end section. These were then stressed together to form a 100m long tube which was then floated out and sunk into position.

Contract	Section	Excavation methods/equipment type	Manufacturer
C104 (shield)	Novena-Newton, Shan Road-Toa Payoh	Shield-mounted Backhoe	Komatsu/Hitachi
C104 (NATM)	Shan Road – Novena	Roadheaders with gathering arms and scraper chain conveyor	Mitsui
C105	Somerset-Orchard-Newton	Tunnel-boring machine	Hitachi Zosen
C106	City Hall-Dhoby Ghaut-Somerset	Shield-mounted rotary backhoe	Grosvenor
C107	Raffles Place-City Hall	Shield-mounted backhoe	Mitsui
C108	Tanjong Pagar-Raffles Place	Shield-mounted boom cutter/backhoe	Mitsubishi
C109	Tiong Bahru-Outram-Tanjong Pagar	Shield-mounted roadheader with scraper chain conveyor	Mitsubishi
C301	City Hall-Bugis	Shield-mounted backhoe/rocker shovel RS95A	Mitsui
C301	Lavender-Bugis	Earth-Pressure Balance Shield	Kawasaki

Fig 1. Tunnelling equipment used on Ph1 & 2

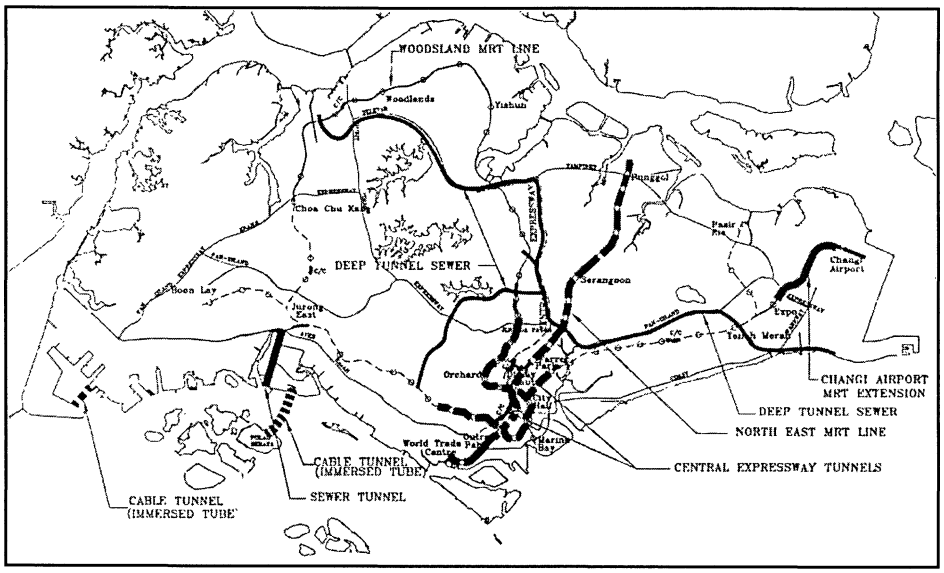


Fig 2. Tunnelling projects in Singapore (completed and under construction)

3.3 *Central Expressway tunnels*

The Central expressway runs roughly north south, from Yio Chu Kang in the north to downtown in the south. It passes through heavily built up areas in the south and thus building the expressway at grade was obviously unacceptable in these locations. Two stretches of the expressway was thus constructed in tunnels. The first 700m long tunnel undercrossing Bukit Timah Road and the second 1.7km long undercrossing Orchard Road and Singapore River. Both of these tunnels were constructed in the late eighties using the cut and cover method. Each direction of traffic is carried in tunnels approximately 20m wide and 8m high accommodating 3 lanes of traffic and an emergency walkway.

4 TUNNELLING/ UNDERGROUND WORKS IN PROGRESS

4.1 *North East line*

This is currently the biggest tunnelling project in Singapore. The whole of the 20km route length runs in tunnels with 16 stations. It was authorised in Jan 1996 and is scheduled for completion in 2002 with a budget of S$5 billion.

The civil engineering works is divided into 12 contract packages and let on a detailed design and build basis. Of these, 3 contracts with no tunnels were for the construction of Dhoby Ghaut Station, Chinatown Station and the Depot. This was due to the scope and complexity of these stations and depot works. All the other 9 contract packages include the construction of stations and tunnels with bored tunnelling specified on 6 of the packages. Tunnels formed by the cut and cover method were mainly used in the outlying areas of Sengkang and Punggol and in areas where 3 parallel tracks are in separate tunnels required to accommodate crossover tracks or emergency sidings along the main line.

The various civil contracts were packaged to ensure that each package is sizeable enough to be attractive to the contractors and at the same time ensure it is not too large so that risks can be contained and managed.

Work on the cut and cover tunnels were completed in June 2000 and the 5.8m diameter twin tunnels lined with precant concrete segments will be completed by October 2000. The tunnelling works were carried out in 24 separate tunnel drives with 16 TBMs. 12 of the TBMs were EPBM, 2 dual mode (EPBM and open mode) and 2 open face shields.

Contact Number	Contractor	Route Length of Twin Tunnel Drives	Type of TBM	Number of TBMs
703, NEL	Samsung Corporation	1286 metres	EPBM -Mitsubishi	2
704, NEL	Wayss & Freytag/Econ Piling/Chew Eu Hock	992 metres plus 1522 metres	EPBM- Lovat	2
705, NEL	Kumagai Gumi/ Sembawang Eng./Mitsui	1349 metres plus 393m/611m	EPBM- Hitachi Zosen	2
706, NEL	Hyundai Eng. & Constr/Zublin AG	910 metres plus 710 metres	EPBM-Herrenknecht	2
708, NEL	Nishimatsu/Lum Chang/Bachy Soletanche	1148 metres plus 402 metres	Semi-Mechanised shield EPBM- Hitachi Zosen	2 2
710, NEL	Shimizu/ Dillingham/Koh Brothers	452 metres plus 2393 metres	EPBM-I.H.I (Japan) EPBM-I.H.I (dual mode)	2 2
503, CAL	Nishimatsu	3500 metres	EPBM- Lovat	2

Fig 3. TBMs on the North East Line (NEL) and Changi Airport Line (CAL)

4.2 *Changi Airport Line*

The other tunnelling works is on the 6.5km extension of the existing West Line from Tanah Merah to the Changi Airport. This line continues from the existing line on viaducts for 2.5 km before going down into 500m long cut and cover section. The remaining length is in twin 5.4 m diameter bored tunnels. One above ground station (Expo Station) is on the viaduct section and the terminal Changi Airport Station is below ground.

Due to the tight programming constraints, the Land Transport Authority, purchased for the first time, by open tender, 2 TBMs ahead of the award of the tunnelling contract. Contractors participating in the tender were given opportunity to price for any modifications that they required to the TBMs specified by the Authority, as the TBMs were to be novated to the successful tenderer.

Tunnelling is mainly in the high abrasive Old Alluvium comprising of weakly cemented sandstone and mudstone that have been subjected to tropical weathering to various depths. Tunnelling todate has been fairly steady with loss time mainly attributable to machine and other downtime of the supporting function. All tunnelling works are expected to be completed by September 2000, with sensitive tunnelling works below the airport runway successfully accomplished.

4.3 *Deep Tunnel Sewerage Scheme*

Phase 1 of the scheme, which is currently being undertaken, is a S$ 3 billion project. When completed it will consist of 50 km of tunnels to intercept existing gravity sewers through drop shafts and channel the flow by gravity flow to a new centralised sewerage treatment works to be commissioned at the southern-eastern end of the island.The treated effluent will be discharged through deep sea outfall pipelines. Phase 1 of the system has been let as 6 design and builds packages.

All the tunnels are tunnels are to be mined with TBMs. Reinforced concrete precast segments provide the initial ground support. Cast-in-situ reinforced concrete provide the secondary lining with its inside surface lined with HDPE or PVC membrane as a primary protection lining. Included in the works are the construction of 10 drop shafts and 28 escape shafts. Works are in the detailed design stage and works on the shaft excavation works have commenced. The first tunnelling work is expected in early 2001.

4.4 *Underground Caverns*

Scarcity of land has been the driving force in looking towards underground caverns for both civilian and military uses. Early studies on the feasibility of the use of caverns were carried out in 1989 and 1993. Further studies were carried out on construction of caverns in Bukit Timah granite since 1996.

Earlier this year the Defence Ministry (MINDEF) announced the construction of an ammunition storage complex in caverns in the Bukit Timah granite at Mandai. This would save more than 300 ha of precious land at ground level. The caverns, each of which can hold up to 100 containers, will be interconnected by tunnels and is expected to be completed by 2003. Construction will be primarily by drill and blast technique at depths of about 100 m and spans up to 60 m.

5 FUTURE PROJECTS

5.1 *Extension of transit line*

The future extension of the transit system will be guided by the necessity of providing inter-suburban connectivity, not well addressed by the basic system, provide bypass route to other parts of the island so that commuters will not have to converge into the busy downtown area to transfer from one line to another and also to connect the sub regional centres.

5.2 *Marina Line*

This 5 km long medium capacity MRT system will be fully underground with 6 stations, running from the existing Dhoby Ghaut station down Bras Basah Road, Nicoll Highway onto the present National Stadium site. Prequalification for civil works for the construction of tunnels and underground stations have been called and the submissions are being reviewed. The main civil contracts, to be divided into 2 contract packages are expected to be called in early 2001. A 5.8 m internal diameter twin tunnel system will be driven for this scheme.

5.3 *Circle Line*

As a system to link the basic E-W line, the N-S line and the North East Line, studies are being undertaken for a possible circle line. This would provide interconnectivity between these lines outside the perimeter of the downtown core as well as between suburban HDB town centres. This line is expected to add another 20 km to 30 km to the transit system with a major portion of it below ground.

A number of feasibility studies are also in hand to serve the eastern residential area, the western industrial area as well as the fast developing Bukit Timah corridor. The long term plans is to extend the transit system to 540 km by year 2030 with an interim target of 270 km by 2010.

5.4 *Deep Tunnel Sewerage Scheme (Phase 2)*

Phase 2 of the DTSS project is expected to be a 17 km tunnel system to serve the western area of Singapore at a budget of S$ 2 billion. The present gravity sewer system in the western area will be intercepted through drop shafts and brought by gravity flow to a coastal treatment works with an outfall into the sea. This Phase is expected to be completed by year 2015.

5.5 *Kallang Expressway/ Paya Lebar Expressway (KLE/ KPE)*

The KLE/ KPE scheme is a 12 km long new expressway linking the north east sector to the city centre. This will provide a much needed relief to the heavily utilised Central Expressway which is currently the only direct expressway link from the city to the north eastern areas. The proposed expressway will also provide a direct link from the city to the rapidly expanding new towns of Sengkang and Punggol.

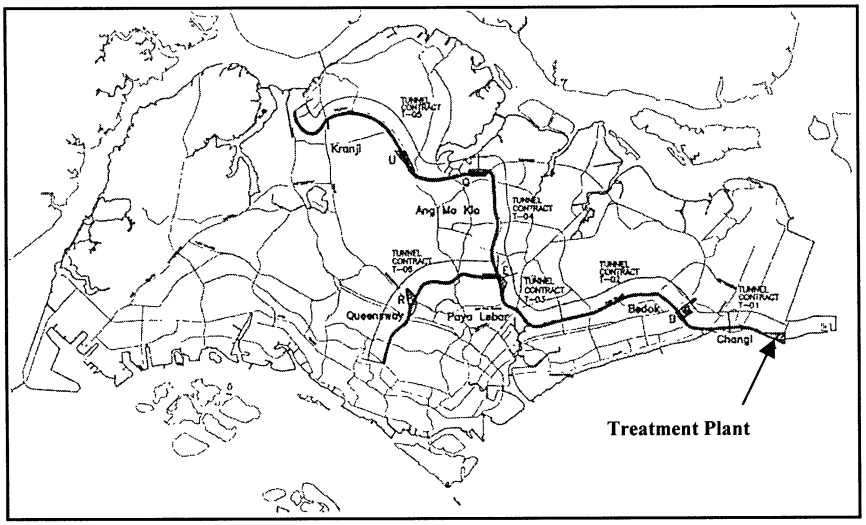


Fig 4. The DTSS Scheme (Phase1)

Contract	Length (m)	Diameter (m)	Dates	Duration (months)	Cost (S$ million)
T01 - Changi Tunnel	5, 791	6.0	Jan00-Dec00	47	80.6
T02 - Bedok Tunnel	7,716	6.0	Dec99-Feb04	50	95.6
T03 – Paya Lebar Tunnel	5,100	6.0	Jan00-Dec03	47	68.8
T04 – Ang Mo Kio Tunnel	7,268	4.3	Feb00-Mar04	49	74.2
T05 – Kranji Tunnel	12,600	3.6	Dec99-Mar04	51	139.5
T06 – Queensway Tunnel	9,600	3.3	Mar00-May04	50	92.0
Total	48,075				550

FIG. 5 Contracts let todate on the DTSS (Phase 1)

The KLE is a part 3 lanes tunnel and part dual 4 lanes depressed expressway with a total route length of 2.8 km linking the East Coast Parkway and the PanIsland Expressway. 2.3 km of it is in depressed road.

The proposed PLE is a part dual 3-lanes tunnel and part dual 3-lanes at grade expressway linking KLE with the Tampines Expressway in the northeast. The total length is 8.8 km of which 5.4 km is in tunnels and 0.3 km as a depressed road.

The 3-lanes tunnel will consist of 2 cell-box structure with each cell approximately 16.5 m wide with a minimum vertical headroom clearence of 5.1 m. The KLE is expected to be completed by 2005 and the PLE by 2006.

5.6 *Future cavern projects*

Geotechnical studies have been carried out between 1996 and 1998 by NTU and the former PWD in sedimentary rocks of the Jurong formation. This is expected to open up possibilities of developing underground space in materials other than the Bukit Timah granite rocks. A significant geology of Singapore is made of sedimentary formation and this would help to open up usable volume without using up valuable surface land area and helping to keep the natural environmental intact.

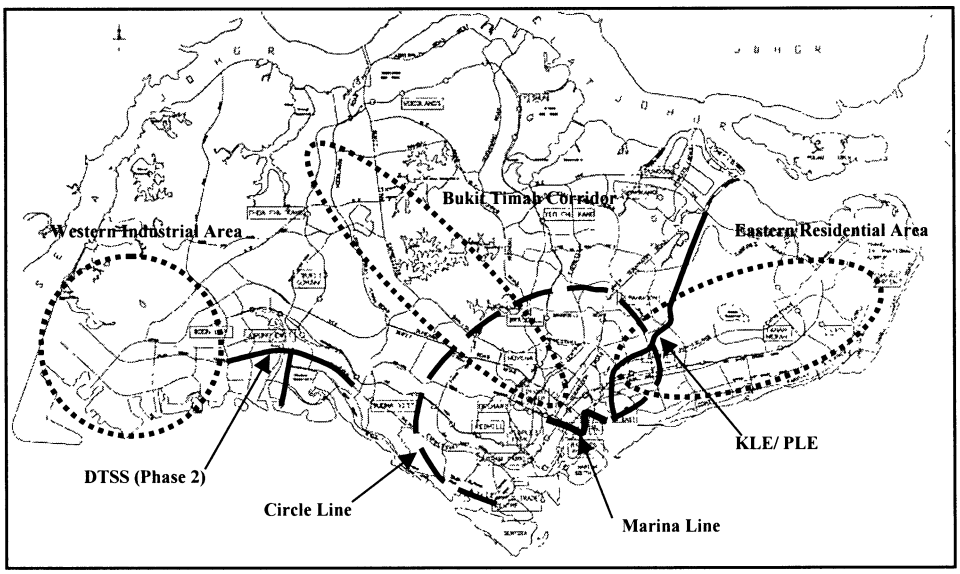


Fig 6. Future Schemes

6 CONCLUSION

Put on an international perspective, the history of tunnelling and underground works in Singapore is relatively young, The first tunnelling works commenced only in 1983. However significant development in underground tunnelling works has taken place since. This development has been made possible by local and international teams of consultants, designers, geotechnical specialist, contractors and tunnelling and specialist equipment suppliers. The rapid growth in this direction in the last 17 years has seen the development of home grown talent in these speacialised fields.

Scarcity of open land and the high price of land in Singapore coupled with ever growing environmental concerns are factors that will drive major infrastructure works below ground. Phase 2 of the deep Tunnel Sewerage Scheme, extension of transit system to 540 km and the planned 80 km long underground road system for the city centre and the possible cavern schemes will be major underground infrastructural investments. Thus the scope of tunnelling and underground works is expected to increase even more rapidly in Singapore in the future years.

REFERENCES

JS Sharma, J. Chu &j. Zhao 1999. Geological and Geotechnical features of Singapore: On overview *Tunnelling & Underground Space Technology* Vol 14 No 4; Pergamon

R. Krishnan, J.P. Copsey, R.G. Algeo & J.N. Shirlaw 1999. Design of Civil Works for Singapore's North East Line. *Tunnelling & Underground Space Technology* Vol 14 No 4; Pergamon

T.W. Hulme, A.J. Burchell 1992. Bored Tunnelling for Singapore Metro. *Journal of Construction and Engineering Management*, ASCE Vol 118 No 2.

T.W. Hulme, R. Krishnan 1998. Singapore MRT: A construction challenge. *Geotechnical special publication No 86. ASCE*